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Cuttings trap for “at bit” measurement of drilled cuttings

Abstract

A cuttings collection system includes a drill bit at the end of a toolstring, a stabilizer connected to the drill bit, a cuttings catcher arranged around the stabilizer with apertures having a size less than an average size of cuttings generated by the drill bit and retractability, and sensors installed in a body of the stabilizer. A cuttings catcher includes a housing with recesses and a retractable mesh installed in each of the recesses. A method of operating an at-bit cuttings catcher system includes providing a toolstring with a drill bit, a stabilizer, a cuttings catcher, and a measurement-while-drilling tool, where the cuttings catcher includes a housing with recesses and a mesh installed in the recesses, drilling a formation using the drill bit, producing cuttings of the formation, extending the mesh from the housing of the cuttings catcher, catching cuttings in the cuttings catcher, and analyzing the cuttings using sensors.

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Background/Summary

BACKGROUND

(1) In drilling operations, harsh drilling conditions, such as low rate of penetration (ROP), vibrations, high pressures and high temperatures, are often present, all of which place increased stress on drill bits. In conventional drilling operations, real-time drill bit information or data is not available. Drilling advisors are required to consider drill bit condition when making decisions to optimize drilling parameters. Further, drill bit condition affects both invisible lost time (ILT) and non-productive time (NPT), which contribute to overall drilling costs. For example, ROP drops as wear progresses (contributing to ILT) and significant wear requires a bit replacement operation (contributing to NPT). Understanding bit conditions in real-time can extend bit life, reducing ILT and NPT.

(2) Cuttings generated during drilling operations provide real-time characteristics of the drilled formation. For example, real-time elemental analysis from x-ray fluorescence (XRF) on drilling cuttings can assist in identifying rock which may cause drill bit metamorphism. The data collected from XRF may be directed to geo-steering users to identify lithological heterogeneity. Further, real-time application of alkene detection for identifying drill bit metamorphism combined with XRF elemental analysis to identify an abrasive silica rich layer can be utilized to aid geo-steering and to provide early alerts to prevent excessive drill bit wear. If necessary, the BHA may be withdrawn from the wellbore to prevent severe damage to the bit, which may assist in avoiding junk in the wellbore.

(3) Currently, manual collection and identification of cuttings is a tedious and unfeasible task for real-time applications. Further, current processes for identification of sub-optimal equipment performance are often unreliable and inconsistent due to unexpected or unplanned solids in rig processing or solid control systems.

SUMMARY

(4) This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

(5) In one aspect, embodiments disclosed herein relate to a cuttings collection system. The cuttings collection system may include a drill bit provided at an end of a toolstring, a stabilizer connected to the drill bit, a cuttings catcher arranged around the stabilizer, and one or more sensors installed in a body of the stabilizer. The cuttings catcher may include apertures having a size less than an average size of cuttings generated by the drill bit and may be retractable.

- (6) In other aspects, embodiments disclosed herein relate to a cuttings catcher including a housing with one or more recesses and a retractable mesh installed in each of the one or more recesses.
- (7) In other aspects, embodiments disclosed herein relate to a method of operating an at-bit cuttings catcher system. The method may include providing a toolstring where the toolstring includes a drill bit, a stabilizer, a cuttings catcher, and a measurement-while-drilling tool, where the cuttings catcher includes a housing with one or more recesses and a mesh installed in one or more recesses. The method may include drilling a formation using the drill bit, producing a plurality of cuttings of the formation, extending the mesh from the housing of the cuttings catcher, catching a plurality of cuttings in the cuttings catcher, and analyzing the plurality of cuttings using one or more sensors.
- (8) Other aspects and advantages of the claimed subject matter will be apparent from the following description and the appended claims.
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Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) Specific embodiments of the disclosed technology will now be described in detail with reference to the accompanying figures. Like elements in the various figures are denoted by like reference numerals for consistency. The size and relative positions of elements in the drawings are not necessarily drawn to scale. For example, the shapes of various elements and angles are not necessarily drawn to scale, and some of these elements may be arbitrarily enlarged and positioned to improve drawing legibility. Further, the particular shapes of the elements as drawn are not necessarily intended to convey any information regarding the actual shape of the particular elements and have been solely selected for ease of recognition in the drawing.
- (2) FIG. 1 shows an exemplary well system.
- (3) FIG. 2 shows an at-bit cuttings collection system in accordance with one or more embodiments.
- (4) FIG. 3 shows a cuttings catcher in accordance with one or more embodiments.
- (5) FIGS. 4A and 4B show a cuttings catcher in accordance with one or more embodiments.
- (6) FIGS. 5A and 5B show a cuttings catcher in accordance with one or more embodiments.
- (7) FIG. 6 shows a flowchart of a method in accordance with one or more embodiments.

DETAILED DESCRIPTION

- (8) In the following detailed description of embodiments of the disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art that the disclosure may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description.
- (9) In the following description of FIGS. 1-5, any component described with regard to a figure, in various embodiments disclosed herein, may be equivalent to one or more like-named components described with regard to any other figure. For brevity, descriptions of these components will not be repeated with regard to each figure. Thus, each and every embodiment of the components of each figure is incorporated by reference and assumed to be optionally present within every other figure having one or more like-named components. Additionally, in accordance with various embodiments disclosed herein, any description of the components of a figure is to be interpreted as an optional embodiment which may be implemented in addition to, in conjunction with, or in place of the embodiments described with regard to a corresponding like-named component in any other figure.
- (10) It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a seismic data set” includes reference to one or more of such seismic data set.
- (11) Terms such as “approximately,” “substantially,” “about,” etc., mean that the recited

characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to those of skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.

(12) It is to be understood that one or more of the steps shown in the flowcharts may be omitted, repeated, and/or performed in a different order than the order shown. Accordingly, the scope disclosed herein should not be considered limited to the specific arrangement of steps shown in the flowcharts.

(13) In one aspect, embodiments disclosed herein relate to an at-bit cuttings collection system secured to a toolstring, where the system includes a cuttings catcher configured to collect cuttings for real-time analysis.

(14) FIG. 1 illustrates an exemplary well system **100** in accordance with one or more embodiments. As shown in FIG. 1, a well path **110** may be drilled by a drill bit **112** attached by a toolstring **104** to a drill rig **102** located on the surface of the earth **106**. The well may traverse a plurality of overburden layers **108** and one or more cap-rock layers **114** to a formation **116**. The well path **110** may be a curved well path, or a straight well path. In one or more embodiments, the well path **110** may be described as vertical, deviated, horizontal, or extended reach drilling (ERD). One skilled in the art will be aware that deviated, horizontal, and ERD wells are considered to be complex.

(15) FIG. 2 shows a cuttings collection system **200** in accordance with one or more embodiments. A near-bit stabilizer **202** may be secured directly uphole of the drill bit **112**. The near-bit stabilizer **202** may be a sleeve-type stabilizer or an integral-type stabilizer. One or more sensors **204** may be secured to the near-bit stabilizer **202**. A cuttings catcher **206** may be secured around the near-bit stabilizer **202**. The cuttings catcher **206** may include mesh halves attached to a housing. A measurement-while-drilling (MWD) tool **208** may be secured to the toolstring uphole of the cuttings catcher **206**.

(16) FIG. 3 shows the cuttings catcher **206** in accordance with one or more embodiments. The cuttings catcher **206** may include a housing **302** which may include a central aperture **304** which may fit around the stabilizer **202**. One or more recesses **306** may be integrally formed in the housing **302**. In one or more embodiments, each of the one or more recesses **306** may have a U-shaped cross-sectional profile, with three straight sides. However, there may be other embodiments in which the one or more recesses **306** have a different shape without departing from the scope of this disclosure. A mesh **308** may be installed in at least one of the recesses **306**. For example, in some embodiments, a mesh **308** may be installed in two of the recesses **306**.

(17) In one or more embodiments, as shown in FIGS. 4A-B, the mesh **308** may be composed of two halves, **308a** and **308b**. Each of the halves **308a**, **308b** may be secured to the housing **302** via one or more connection points **402** integrally formed in each half **308a**, **308b** and via one or more pivots **404** integrally formed in the recess wall **406** of the housing **302**. Each of the connection point **402** and the pivot **404** may represent an extended member with a central aperture through which a shaft **408** may be fitted. The mesh halves **308a**, **308b** may therefore rotate about a central axis **410** of the shaft **408** between a retracted position and an extended position, which are illustrated in FIGS. 5A and 5B, respectively. In one or more embodiments, the mesh halves **308a**, **308b** may extend and retract independently of each other. Additionally, in one or more embodiments, other hinge-type connection configurations may be used to pivotably connect a mesh within a recess.

(18) As shown in FIG. 5A, when the mesh halves **308a**, **308b** are in a retracted position, they may be flush with the recess wall **406**. A retracted position, therefore, may refer to the position of the mesh halves **308a**, **308b** when they are at 0° rotation about the central axis **410** of the shaft **408**. An extended position, shown in FIG. 5B, may refer to the position of the mesh halves **308a**, **308b** when they are at 90° rotation about the central axis **410** of the shaft **408**. Accordingly, in an extended position, the mesh halves **308a**, **308b** may fill the recess **306**, creating a barrier through

which cuttings cannot pass. The size of the mesh **308** apertures may be selected by a user to accommodate the expected cuttings size. For example, the size of the mesh apertures may be selected to be less than an average size of cuttings from the connected drill bit **112**. The average size of cuttings generated from a drill bit may be predicted (e.g., from previously collected data from drilling like-type formations) and/or determined from cuttings collection at the surface, for example.

(19) The mesh halves **308a**, **308b** may be rotated from a 0° rotation to a 90° rotation. The rotation may be actuated by a hydraulic motor. The hydraulic motor may be any commercially available hydraulic motor suitable for use downhole.

(20) When the mesh **308** is in an extended position, the mesh **308** may allow for the collection of cuttings. Once cuttings have been collected, the sensors **204**, shown in FIG. 2, may be used to analyze the cuttings. For example, the sensors **204** may be configured to measure various formation characteristics, such as temperature, pressure, force, gamma-rays, and combinations thereof. Measurements taken by the sensors **204** may be sent (e.g., via wired connection) to an MWD tool **208** secured to the toolstring uphole of the cuttings catcher **206**. In some embodiments, measurements taken by the sensors **204** may be sent through a wired connection to the surface of the well, for example, to a computer system located at the surface, for data collection and analysis. Once the desired measurements have been made, the mesh **308** may then be retracted, releasing the cuttings into the wellbore.

(21) FIG. 6 depicts a flowchart in accordance with one or more embodiments. More specifically, FIG. 6 depicts a flowchart **600** of a method of determining successful functionality of one or more inflow control devices according to embodiments of the present disclosure. Further, one or more blocks in FIG. 6 may be performed by one or more components as described in FIGS. 1-5B. While the various blocks in FIG. 6 are presented and described sequentially, one of ordinary skill in the art will appreciate that some or all of the blocks may be executed in different orders, may be combined, may be omitted, and some or all of the blocks may be executed in parallel. Furthermore, the blocks may be performed actively or passively.

(22) Initially, a toolstring **104** may be provided, **S602**. The toolstring **104** may include a drill bit **112**, a stabilizer **202**, a cuttings catcher **206** and a MWD tool **208**. The cuttings catcher **206** may include a housing **302** with one or more recesses **306** and at least one mesh **308** installed in at least one of the recesses **306**. The mesh **308** may be composed of two halves **308a**, **308b**, each of which may be configured to retract and extend as desired, either in unison or independently of one another.

(23) The formation **116** may be drilled using the drill bit **112**, **S604**. In the process of drilling, a plurality of cuttings of the formation **116** may be generated, **S606**. The size of the cuttings, in accordance with one or more embodiments, may be estimated, and the estimation may be used to select the size of the mesh **308** installed in the cuttings catcher **206** so that cuttings are not able to pass through apertures in the mesh **308**.

(24) The mesh **308** may extended from the housing **302** of the cuttings catcher **206**, **S608**. More specifically, the mesh halves **308a**, **308b** may be rotated 90° from a retracted position to an extended position about the central axis **410** of the shaft **408**. After extension of the mesh halves **308a**, **308b**, the plurality of cuttings may be caught in the cuttings catcher **206**, **S610**. Once caught, the cuttings may be analyzed using sensors **204** installed on the stabilizer **202**, **S612**.

(25) In one or more embodiments, once the desired measurements and analysis of the cuttings has been completed, the mesh **308** may be retracted into the housing **302** of the cuttings catcher **206**. Once the mesh **308** has been retracted, the plurality of cuttings may then be released into the wellbore.

(26) In one or more embodiments, extension and retraction of the mesh **308** may be controlled by a user at a surface location, who may manually input a user command to retract or extend the mesh **308** via a wired connection through the MWD tool **208**. In other embodiments, extension and

retraction of the mesh **308** may be automatically controlled based on a predetermined timed schedule. More specifically, extension and retraction of the mesh **308** may occur in intervals after a desired period of time. For example, in one or more embodiments, the mesh **308** may be extended and remain extended for short period of time, such as one minute, before it may be retracted and remain retracted for a longer period of time, on the order of a few hours. This may be repeated throughout the drilling operation to acquire regular cuttings analysis.

(27) Catching and analysis of formation **116** cuttings proximate the drill bit **112** allows for real-time determination of formation **116** characteristics, which has previously been unfeasible. Accordingly, drilling conditions, particularly those related to wear and condition of the drill bit **112**, may be transmitted to users in real-time, so that decisions may be made in relation to preservation of drilling equipment. Further, predictions regarding the reservoir and its potential may also be made by users in real-time due to measurements taken by the sensors **204** installed proximate the cuttings catcher **206**.

(28) Embodiments of the present disclosure may provide at least one of the following advantages. Conventionally available drilling systems are not capable of providing real-time data regarding drill bit condition and formation characteristics. Rather, commercially available systems often require drill bit replacement due to bit damage, which results in costly time delays and reduction in drilling efficiency. Embodiments of the present disclosure allow for real-time, near-bit analysis of formation cuttings to the surface for the purpose of monitoring drill bit condition and formation characteristics. Such real-time data allows operators to make more informed decisions regarding drilling operations, particularly with respect to drill bit replacement. Real-time data allows for a reduction in non-production time and invisible lost time, since it allows users to determine when the condition of the drill bit has deteriorated to a level at which drilling efficiency is no longer above an acceptable threshold.

(29) Although only a few example embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the example embodiments without materially departing from this invention. Accordingly, all such modifications are intended to be included within the scope of this disclosure as defined in the following claims. Although multiple dependent claims are not introduced, it would be apparent to one of ordinary skill that the subject matter of the dependent claims of one or more embodiments may be combined with other dependent claims.

Claims

1. A cuttings collection system, comprising: a drill bit provided at an end of a toolstring; a stabilizer connected to the drill bit; a cuttings catcher secured to the stabilizer, the cuttings catcher comprising: a housing having a recess; and a mesh connected to the recess via a pivot connection, wherein the mesh comprises apertures having a size less than an average size of cuttings generated by the drill bit, and wherein the mesh is retractable; and one or more sensors installed in a body of the stabilizer.
2. The cuttings collection system of claim 1, wherein the one or more sensors are configured to analyze a plurality of cuttings.
3. The cuttings collection system of claim 1, wherein the housing comprises one or more additional recesses, and wherein an additional mesh is installed in each of the one or more additional recesses, wherein the apertures are formed through the additional mesh.
4. The cuttings collection system of claim 1, wherein the pivot connection comprises a pivot integrally formed in the recess of the housing, and a shaft extending through the pivot and a connection point on the mesh.
5. The cuttings collection system of claim 4, wherein the mesh is configured to rotate about the shaft between 0° and 90°, wherein 0° represents a retracted position and 90° represents an extended

position.

6. The cuttings collection system of claim 1, wherein the recess has a U-shaped cross-sectional profile.

7. The cuttings collection system of claim 1, wherein the mesh comprises two halves, each half configured to independently retract.

8. The cuttings collection system of claim 1, further comprising a measurement-while-drilling (MWD) tool wired uphole of the stabilizer.

9. A cuttings catcher, comprising: a housing comprising a recess; a retractable mesh installed in the recess; and a pivot integrally formed in the one or more recesses of the housing, and a shaft extending through the pivot and a connection point on the retractable mesh, wherein, when the retractable mesh is in an extended position, the retractable mesh extends across the recess to fill the recess.

10. The cuttings catcher of claim 9, wherein the retractable mesh is configured to rotate about the shaft between 0° and 90°, wherein at 0°, the retractable mesh is in a retracted position, and at 90°, the retractable mesh is in the extended position.

11. The cuttings catcher of claim 9, wherein the recess has a U-shaped cross-sectional profile.

12. The cuttings catcher of claim 9, wherein the retractable mesh comprises two halves, each half configured to independently retract.

13. A method of operating an at-bit cuttings catcher system, comprising: providing a toolstring, the toolstring comprising a drill bit, a stabilizer, a cuttings catcher, and a measurement-while-drilling (MWD) tool, wherein the cuttings catcher is secured to the stabilizer, and wherein the cuttings catcher comprises a housing with one or more recesses, and a mesh installed in the one or more recesses; drilling a formation using the drill bit; producing a plurality of cuttings of the formation; extending the mesh from the housing of the cuttings catcher, wherein extending the mesh from the housing of the cuttings catcher comprises rotating the mesh about a shaft at a connection point from a retracted position to an extended position; catching a plurality of cuttings in the cuttings catcher; and analyzing the plurality of cuttings using one or more sensors.

14. The method of claim 13, further comprising: retracting the mesh into the housing of the cuttings catcher; and releasing the plurality of cuttings.

15. The method of claim 14, wherein retracting the mesh into the housing of the cuttings catcher comprises rotating the mesh about a shaft at a connection point from an extended position to a retracted position.

16. The method of claim 13, further comprising controlling an extension and a retraction of the mesh with a user command inputted at a surface location.

17. The method of claim 13, further comprising automatically controlling an extension and a retraction of the mesh on a timed schedule.

18. The method of claim 13, wherein analyzing the plurality of cuttings comprises collecting data from the one or more sensors using the measurement-while-drilling (MWD) tool and transmitting the data via a wired connection to a user at a surface location.
